

a Task-specific method-selection guide

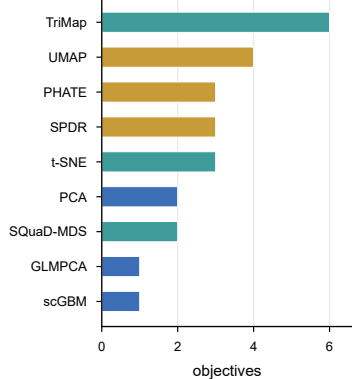
	Method	Structure and clustering			Runtime			Memory			Stability	
		Local	Global	Cluster	5k	20k	50k	5k	20k	50k	Dropout	Batch
Rare types	TriMap	±	±	±	5s	17s	41s	353M	635M	1.2G	+	+
	SQuaD-MDS	±	+	-	49s	4m	10m	370M	682M	1.4G	-	-
	UMAP	±	±	±	18s	21s	33s	924M	1.3G	2.3G	+	+
	SPDR	±	±	±	2m	22m	>2h	715M	1.9G	4.3G	+	+
	TriMap	±	±	±	5s	17s	41s	353M	635M	1.2G	±	±
Trajectory	PCA	±	+	±	2s	5s	9s	267M	824M	2.2G	±	±
	SQuaD-MDS	±	+	-	49s	4m	10m	370M	682M	1.4G	-	-
	PHATE	±	±	±	19s	51s	4m	516M	851M	1.6G	±	±
Clustering	scGBM	±	±	±	38s	2m	6m	605M	1.6G	3.4G	±	±
	TriMap	±	±	±	5s	17s	41s	353M	635M	1.2G	+	+
	t-SNE	±	±	±	14s	1m	4m	272M	770M	1.9G	+	+
Large scale	GLMPCA	±	±	±	16s	2m	7m	454M	1.4G	3.7G	±	±
	PCA	±	+	±	2s	5s	9s	267M	824M	2.2G	±	±
	TriMap	±	±	±	5s	17s	41s	353M	635M	1.2G	+	+
Noise	UMAP	±	±	±	18s	21s	33s	924M	1.3G	2.3G	+	+
	PHATE	±	±	±	19s	51s	4m	516M	851M	1.6G	±	±
	TriMap	±	±	±	5s	17s	41s	353M	635M	1.2G	+	+
Balanced	SPDR	±	±	±	2m	22m	>2h	715M	1.9G	4.3G	+	+
	t-SNE	±	±	±	14s	1m	4m	272M	770M	1.9G	+	+
	UMAP	±	±	±	18s	21s	33s	924M	1.3G	2.3G	+	+
	TriMap	±	±	±	5s	17s	41s	353M	635M	1.2G	+	+
	PHATE	±	±	±	19s	51s	4m	516M	851M	1.6G	±	±
	SPDR	±	±	±	2m	22m	>2h	715M	1.9G	4.3G	+	+

Good/Fast

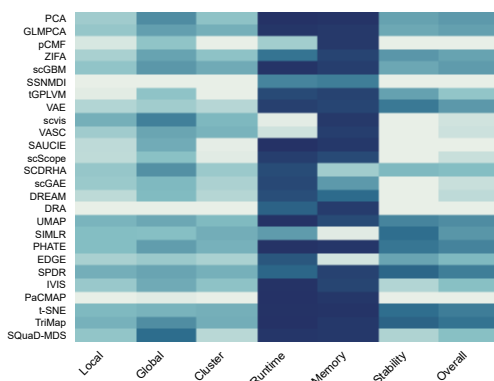
Medium

Poor/Slow

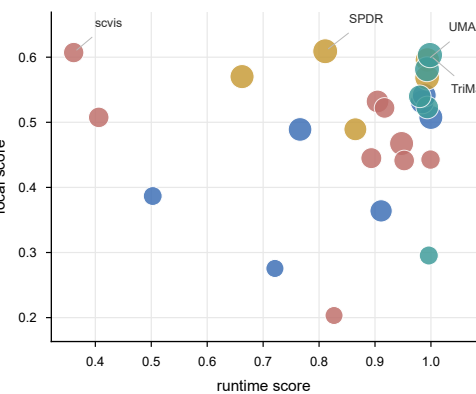
b Repeated recommendations



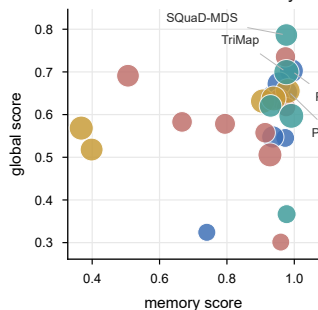
c Benchmark scores



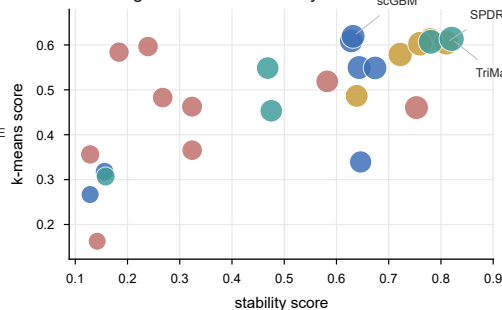
d Local score versus runtime



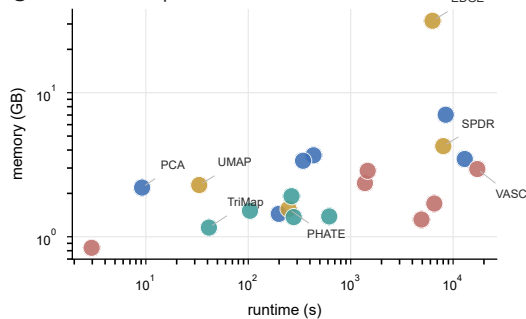
e Global score versus memory



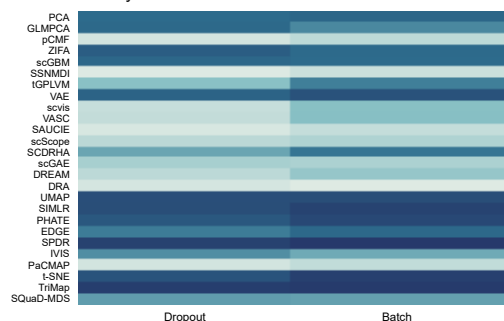
f Clustering score versus stability



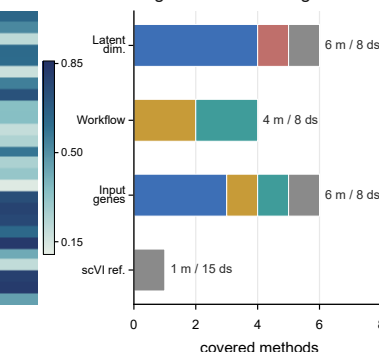
g 50k-cell footprint



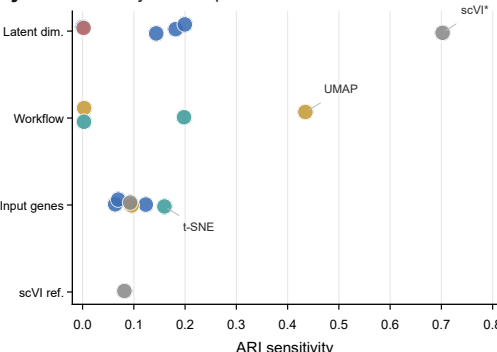
h Stability scores



i Targeted-control design



j Sensitivity landscape



Method family
 linear deep graph metric targeted control